Functional Requirements Document – Mail

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (Sub-module: Communication → Mail, code `MAIL`) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** BRD-Mail.md v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related Documents

Document	Purpose	Location
BRD – Mail	Business needs this FRD details (`REQ-MAIL-nn`, `BR-MAIL-nn`, `AC-MAIL-nn`)	`../../../BRD/04-Communication/04.1-Mail/BRD-Mail.md`
FRD – Overall	Roles, global rules, Authentication, Legal, module index	`../../00-Overall/FRD-TaskFlow-Overall.md`
FRD – Notification (sibling)	Reminder engine that places reminder mails in the Inbox	`../04.2-Notification/FRD-Notification.md`
FRD – Reminder Settings	"Also deliver reminders to Mail" toggle	`../../06-Settings/06.1-Reminder-Settings/FRD-Reminder-Settings.md`
FRD – Task Management	Comments, reassignment and worked hours that produce mails	`../../02-TaskFlow-Monitoring/02.1-Task-Management/FRD-Task-Management.md`
FRD – Assistant	Finding users and drafting a mail through the assistant	`../../08-TFA-Assistant/08.1-Assistant/FRD-Assistant.md`
Development Plan	Screen 14 (marked SUPERSEDED by internal Mail)	`TF - Documents/Screens Wise Feature Development Plan/14-email-notification-settings-screen.md`
HLD / LLD	Technical design of Mail	`../../../HLD/04-Communication/04.1-Mail/HLD-Mail.md`, `../../../LLD/04-Communication/04.1-Mail/LLD-Mail.md`

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Mail sub-module: the Mail screen (folders, list, reading pane, organising actions, attachment downloads), the compose panel (new mail, reply, reply all, forward, drafts, attachments) and the system mails that other sub-modules place in a user's Inbox.

1.2 Scope

In scope: everything listed in BRD-Mail §3.1. Out of scope (BRD-Mail §3.2): outside email (SMTP) and its settings, reading another person's mailbox, rich-text bodies, physical clean-up of deleted mail and files, and the generation of deadline reminders (specified in the Notification FRD; this FRD covers only their arrival in the Inbox).

Change from original BRD

Original BRD item	Current requirement
Screen 14 "Email Configuration" (SMTP server, port, sender, credentials, test email)	Superseded and not built; replaced by the internal Mail screen specified here (FR-MAIL-01 – FR-MAIL-56).
Screen 14 "Enable/disable email notifications"	The "Also deliver reminders to Mail" toggle in Reminder Settings decides whether reminders and comment alerts reach the Inbox (FR-MAIL-25).
Screen 14 FR 5 "Send deadline reminders"	Reminders arrive in the Inbox as mails from "TaskFlow" (§5.1; generated by the Notification sub-module).
Screen 14 FR 6 / FR 7 (delivery failures, fallback to in-app)	No outside mail service; the in-app notification is always created and the Inbox mail is an extra copy (open issue in §10).

1.3 Audience

Developer, tester, reviewer and product owner of TaskFlow.

1.4 Conventions

- Requirement IDs: `FR-MAIL-nn`. Use cases: `UC-MAIL-nn`. BRD items: `REQ-MAIL-nn`, sub-module rules `BR-MAIL-nn`, global rules `BR-nn` (overall BRD), NFRs `NFR-MAIL-nn`, acceptance criteria `AC-MAIL-nn`.
- Priority: **Must / Should / Could** (MoSCoW), taken from the traced BRD requirement.
- "The system shall …" = mandatory behaviour. Text in *italics* inside messages is a value filled in at run time.

1.5 Definitions & Abbreviations

Term	Meaning
Mailbox	A user's own folders and mail copies; scope is always **Own**
Mail copy	One user's entry for a mail in a folder (Inbox, Sent, Drafts) with personal read, star, important and trash state
Conversation	A mail and all its replies and forwards, shown together
Draft	A saved, unsent mail in the Drafts folder
System mail	A mail sent by TaskFlow itself, shown with the sender "TaskFlow" (reminders, comment alerts)
Pending upload	A file uploaded in the compose panel that is not yet part of a sent mail or saved draft
Delete forever	Removes a mail copy from the user's view permanently; only from Trash or for a draft
Active user	A TaskFlow user account that is not deactivated or deleted

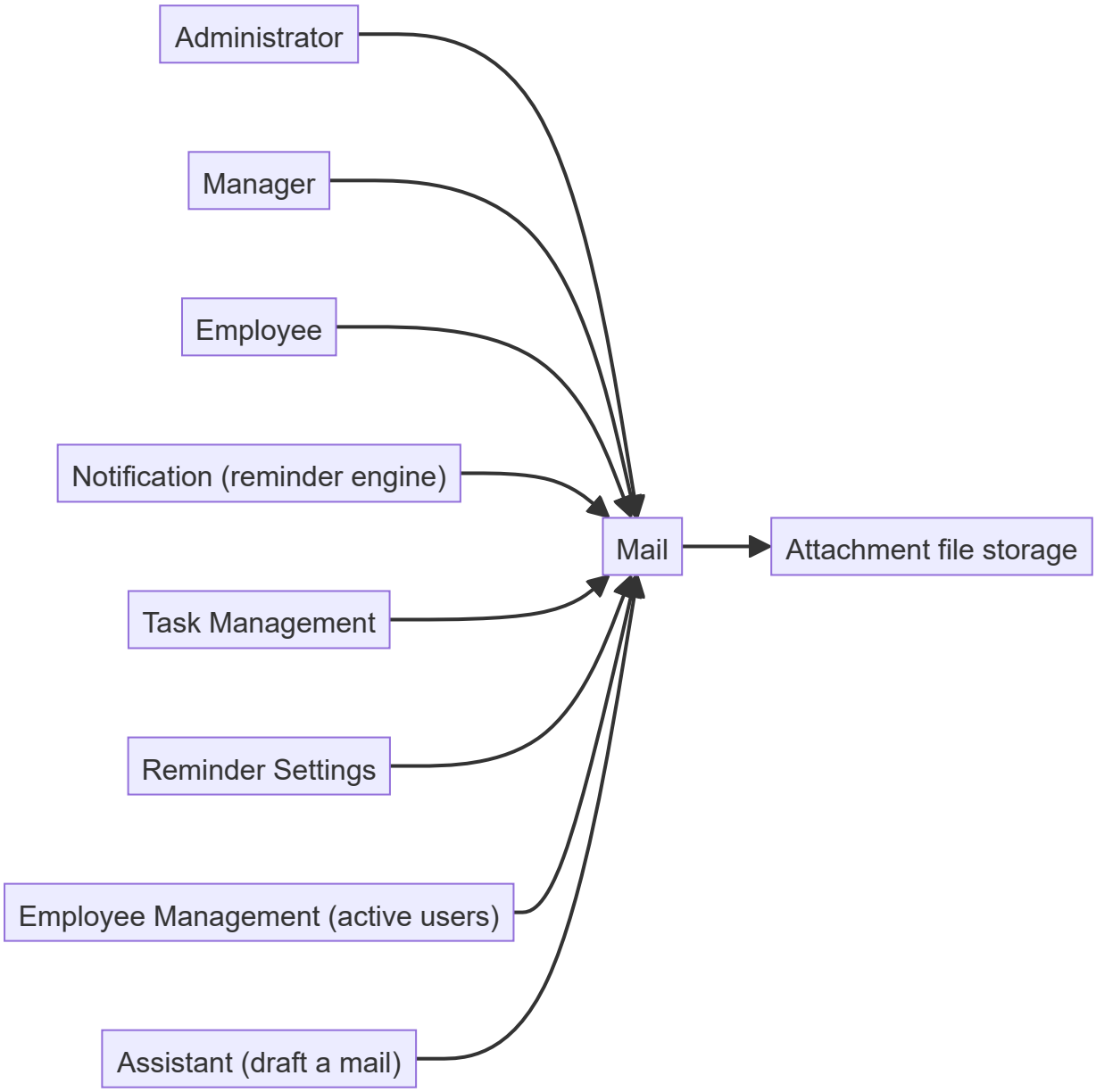
2. System Overview

2.1 Module Map

Code	Module	Child modules	Primary roles
MAIL	Communication → Mail	Mailbox (Mail screen), Compose panel, Attachments, System mails received	Administrator, Manager, Employee, Custom Role (as granted)

Other modules are described in the overall FRD (`../../00-Overall/FRD-TaskFlow-Overall.md`).

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	System administrator; also grants `mail.*` keys to custom roles in Role Configuration	Own mailbox; recipient suggestions over all active users
Manager	Leads a team or project	Own mailbox; recipient suggestions over all active users
Employee	Works on assigned tasks	Own mailbox; recipient suggestions over all active users
Custom Role	Role created by an Administrator	Own mailbox, as granted

No role, including Administrator, can read another person's mailbox (BR-MAIL-02).

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Mail › Open Mail, read folders and mails, menu badge	`mail.read`	Granted	Granted	Granted	Configurable
Mail › Compose, reply, forward, drafts, recipient suggestions	`mail.compose`	Granted	Granted	Granted	Configurable
Mail › Star, important, unread, trash, restore, delete forever	`mail.organize`	Granted	Granted	Granted	Configurable
Mail › Upload, remove and download attachments	`mail.attachments`	Granted	Granted	Granted	Configurable
Reminder Settings › "Also deliver reminders to Mail" (controls system mails; specified in the Reminder Settings FRD)	`settings.reminders.manage`	see Reminder Settings FRD			

A new custom role starts with none of the four `mail.*` keys.

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4. Functional Requirements by Module

4.1 Module: Mail (MAIL)

Purpose: A private internal mailbox for messages between TaskFlow users, with system mails about tasks in the same Inbox. · **BRD reference:** BRD-Mail §5 SCREEN 1, SCREEN 2; BR-MAIL-01 – BR-MAIL-13; global BR on private messages.

4.1.1 Screen: Mail

Field	Value
Route / location	`/mail` (menu: Communication Management → Mail, with unread badge)
Access	`mail.read` (Administrator, Manager, Employee by default; Custom Role if granted); actions need `mail.organize`, `mail.compose` or `mail.attachments`

Field	Value
Entry points	Side menu item "Mail"; "Open the task" link leads out to the task

Screen elements

Element	Type	Description	Mandatory	Visible to
Folder rail	Navigation list	Inbox, Starred, Important, Sent, Drafts, Trash with counts	–	`mail.read`
Compose	Button	Opens the compose panel "New mail"	–	`mail.compose`
Search box	Text input	"Search inbox" (label follows the folder); searches the current folder	No	`mail.read`
Message list	List	Rows with sender / "To: …" / "TaskFlow", subject or "(no subject)", preview, date, important flag, paperclip	–	`mail.read`
Star toggle	Icon button	Stars / un-stars the row	–	`mail.organize`
Previous / Next	Buttons	Page through the list, 25 per page	–	`mail.read`
Reading pane	Panel	Subject, "Open the task" link, conversation (sender, recipients, Cc, date, message, attachments)	–	`mail.read`
Attachment buttons	Buttons	Download a file	–	Shown to every reader; use needs `mail.attachments`
Reply / Reply all	Buttons	Pre-filled compose panel; hidden for system mail and Trash	–	`mail.compose`
Forward	Button	Pre-filled compose panel	–	`mail.compose`
Star / Mark important / Mark unread / Move to Trash	Buttons	Personal flags; Mark unread in Inbox only	–	`mail.organize`
Restore / Delete forever	Buttons	Shown in Trash	–	`mail.organize`
Folder button	Button	Opens the folder rail on narrow screens	–	`mail.read`

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-MAIL-01	The system shall open `/mail` only for a user holding `mail.read` and shall return any other user to their dashboard without a message.	Must	`mail.read`	REQ-MAIL-01
FR-MAIL-02	The system shall show the folders Inbox, Starred, Important, Sent, Drafts and Trash and shall open Inbox when the screen is opened.	Must	`mail.read`	REQ-MAIL-02
FR-MAIL-03	The system shall show the number of unread mails beside Inbox, Starred and Important, and the total number of drafts beside Drafts, and shall update them after every read, flag, trash, send or draft action.	Should	`mail.read`	REQ-MAIL-03

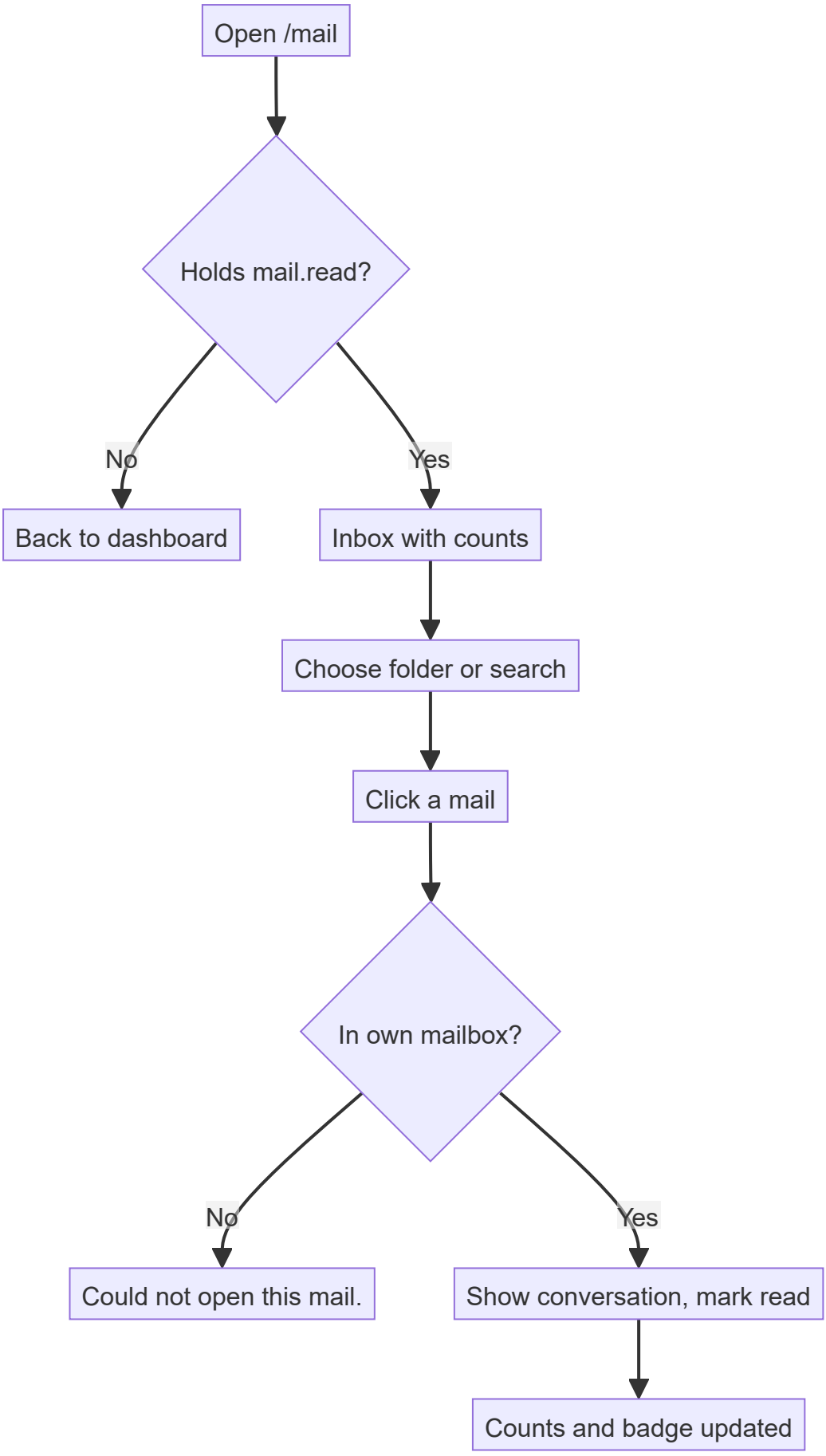
ID	Requirement	Priority	Permission	BRD trace
FR-MAIL-04	The system shall show on the Mail menu item a badge with the number of unread Inbox mails, refreshed on every page change and every 15 seconds, and hidden when the number is 0.	Should	`mail.read`	REQ-MAIL-04, NFR-MAIL-07
FR-MAIL-05	The system shall list the mails of the selected folder newest first, 25 per page, with Previous and Next.	Must	`mail.read`	REQ-MAIL-05, NFR-MAIL-01
FR-MAIL-06	The system shall show on each row the sender (in Sent and Drafts "To: …" followed by the recipients; "TaskFlow" for a system mail), the subject or "(no subject)", a preview of the message, the date, an important flag when set and a paperclip when the mail has attachments.	Must	`mail.read`	REQ-MAIL-06
FR-MAIL-07	The system shall search the current folder for the typed text in the subject, message text, sender and recipients, and restart the list at page 1.	Should	`mail.read`	REQ-MAIL-07
FR-MAIL-08	The system shall show "Nothing in *folder*." for an empty folder, "No mail matches your search." when a search finds nothing, and "Select a mail to read it." while no mail is open.	Should	`mail.read`	REQ-MAIL-05, REQ-MAIL-07
FR-MAIL-09	The system shall show, for an opened mail, the whole conversation oldest first, each mail with sender, To, Cc, date, message and attachments.	Must	`mail.read`	REQ-MAIL-08
FR-MAIL-10	The system shall mark the reader's copy of a mail as read when the reader opens it.	Must	`mail.read`	REQ-MAIL-09
FR-MAIL-11	The system shall show message text as plain text and make only web links starting with http or https clickable.	Could	`mail.read`	REQ-MAIL-10, NFR-MAIL-03
FR-MAIL-12	The system shall show an "Open the task" link on a mail that is about a task, opening that task's details.	Should	`mail.read`	REQ-MAIL-11
FR-MAIL-13	The system shall let a user star and un-star a mail from the list row and from the reading pane.	Should	`mail.organize`	REQ-MAIL-12
FR-MAIL-14	The system shall let a user mark and unmark a mail as important from the reading pane.	Should	`mail.organize`	REQ-MAIL-13
FR-MAIL-15	The system shall offer Mark unread only for mails in the Inbox and shall then show the mail as unread and raise the counts.	Should	`mail.organize`	REQ-MAIL-14
FR-MAIL-16	The system shall move a mail to Trash on Move to Trash and show "Moved to Trash".	Must	`mail.organize`	REQ-MAIL-15
FR-MAIL-17	The system shall return a mail from Trash to its folder on Restore and show "Restored".	Must	`mail.organize`	REQ-MAIL-15
FR-MAIL-18	The system shall delete forever only a mail that is in Trash, or a draft, show "Deleted forever", and refuse any other mail with "Move the mail to Trash before deleting it forever."	Must	`mail.organize`	REQ-MAIL-16, BR-MAIL-04

ID	Requirement	Priority	Permission	BRD trace
FR-MAIL-19	The system shall apply read state, star, important, trash, restore and delete forever only to the acting user's own copy; other people's copies of the same mail stay unchanged.	Must	`mail.organize`	REQ-MAIL-17, BR-MAIL-03
FR-MAIL-20	The system shall never list, open or change a mail that is not in the caller's mailbox and shall treat such a mail as not found ("Could not open this mail." / "Could not update the mail.").	Must	`mail.read`, `mail.organize`	REQ-MAIL-18, BR-MAIL-02, NFR-MAIL-02
FR-MAIL-21	The system shall show a system mail with the sender "TaskFlow" and offer only Forward for it (no Reply, no Reply all).	Must	`mail.compose`	REQ-MAIL-19, BR-MAIL-05
FR-MAIL-22	The system shall not offer Reply or Reply all for a mail in Trash.	Should	`mail.compose`	REQ-MAIL-20
FR-MAIL-23	The system shall keep showing the name of a sender or recipient whose account was later deactivated or deleted.	Should	`mail.read`	REQ-MAIL-21, BR-MAIL-12
FR-MAIL-24	The system shall show one pane at a time (folders, list or reading pane) on narrow screens, with a folder button to open the folder rail.	Could	`mail.read`	REQ-MAIL-22, NFR-MAIL-05
FR-MAIL-25	The system shall, while "Also deliver reminders to Mail" is on, place an Inbox mail from "TaskFlow" with the subject "Comment *verb*: *title*" in the mailbox of the task's assignee, recipients and the acting person whenever a comment is added, edited or deleted.	Should	System (no key)	REQ-MAIL-23, BR-MAIL-13
FR-MAIL-26	The system shall, when a task is reassigned, send a mail with the subject "Task reassigned: *title*" from the person who reassigned it to the task's active participants.	Should	System (no key)	REQ-MAIL-24, BR-MAIL-13
FR-MAIL-27	The system shall, when the worked hours of a task change, send a mail with the subject "Worked hours changed: *title*" from the person who changed them to the task's active assignee and recipients.	Could	System (no key)	REQ-MAIL-25, BR-MAIL-13
FR-MAIL-28	The system shall let a user download an attachment only when the user uploaded it or the mail is in the user's own (not deleted-forever) mailbox, and otherwise report "Could not download the file."	Must	`mail.attachments`	REQ-MAIL-26, BR-MAIL-11
FR-MAIL-29	The system shall always deliver an attachment as a file download and never display it inside the page.	Must	`mail.attachments`	REQ-MAIL-27, NFR-MAIL-03
FR-MAIL-30	The system shall show "Could not load your mail.", "Could not open this mail.", "Could not update the mail." or "Could not mark as unread." when loading, opening or changing a mail fails, including when the user lacks the needed permission.	Should	`mail.read`, `mail.organize`	REQ-MAIL-08, REQ-MAIL-12 – REQ-MAIL-15

Use case: UC-MAIL-01 – Read a mail

Item	Detail
Actor	Any user (Administrator, Manager, Employee, Custom Role with `mail.read`)
Preconditions	Signed in; holds `mail.read`

Item	Detail
Trigger	Opens Communication Management → Mail
Main flow	1. System opens Inbox with folder counts (FR-MAIL-02, 03). 2. User picks a folder or types in the search box (FR-MAIL-05, 07). 3. User clicks a row. 4. System shows the conversation oldest first and marks the copy read (FR-MAIL-09, 10). 5. Counts and the menu badge drop (FR-MAIL-03, 04).
Alternate flows	A1: mail is about a task → user clicks "Open the task" (FR-MAIL-12). A2: narrow screen → one pane at a time (FR-MAIL-24).
Exceptions	E1: no `mail.read` → returned to the dashboard (FR-MAIL-01). E2: mail not in the user's mailbox → "Could not open this mail." (FR-MAIL-20). E3: empty folder / no hits → FR-MAIL-08 texts.
Postconditions	The user's copy is read; nobody else's copy changed

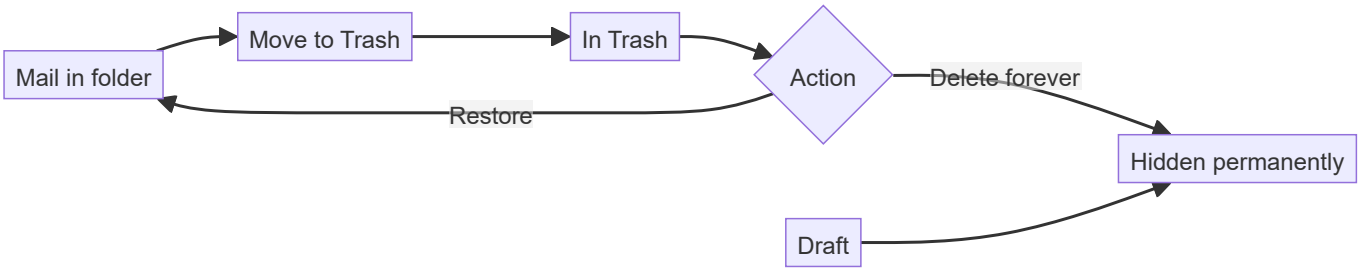


Use case: UC-MAIL-02 – Flag a mail (star, important, unread)

Item	Detail
Actor	User with `mail.organize`
Preconditions	Mail screen open; mail in own mailbox
Trigger	Clicks the star on a row, or Star / Mark important / Mark unread in the reading pane
Main flow	1. System saves the flag on the user's own copy (FR-MAIL-13 – 15, 19). 2. The mail appears in or leaves Starred / Important. 3. Counts update.
Alternate flows	A1: Mark unread is offered only in Inbox (FR-MAIL-15).
Exceptions	E1: no `mail.organize` or foreign mail → "Could not update the mail." / "Could not mark as unread." (FR-MAIL-30).
Postconditions	Flag stored for this user only

Use case: UC-MAIL-03 – Trash, restore or delete forever

Item	Detail
Actor	User with `mail.organize`
Preconditions	Mail in own mailbox
Trigger	Clicks Move to Trash; in Trash clicks Restore or Delete forever
Main flow	1. Move to Trash → "Moved to Trash" (FR-MAIL-16). 2. In Trash, Restore → "Restored" (FR-MAIL-17). 3. In Trash, Delete forever → "Deleted forever"; the mail no longer appears in any folder (FR-MAIL-18).
Alternate flows	A1: a draft can be deleted forever without passing through Trash (FR-MAIL-18).
Exceptions	E1: Delete forever on a mail not in Trash and not a draft → "Move the mail to Trash before deleting it forever."
Postconditions	Only the user's own copy changed (FR-MAIL-19)



Use case: UC-MAIL-04 – Download an attachment

Item	Detail
Actor	User with `mail.attachments`
Preconditions	Mail with attachments open in the reading pane
Trigger	Clicks an attachment button
Main flow	1. System checks that the user uploaded the file or holds the mail in their mailbox (FR-MAIL-28). 2. The browser downloads the file under its original name (FR-MAIL-29).

Item	Detail
Alternate flows	–
Exceptions	E1: no `mail.attachments`, not allowed or file missing → "Could not download the file."
Postconditions	File saved on the user's device; nothing changes in TaskFlow

Use case: UC-MAIL-05 – Receive a system mail

Item	Detail
Actor	System (Notification, Task Management); recipient
Preconditions	Recipient is an active user
Trigger	A reminder is delivered, a comment is added / edited / deleted, a task is reassigned or worked hours change
Main flow	1. Reminder or comment and "Also deliver reminders to Mail" is on → Inbox mail from "TaskFlow" (FR-MAIL-25, §5.1). 2. Reassignment or worked hours → mail from the acting person (FR-MAIL-26, 27). 3. Recipient opens the mail; a task mail shows "Open the task" (FR-MAIL-12).
Alternate flows	A1: toggle off → no reminder or comment mail; the in-app notification is still created (Notification FRD).
Exceptions	E1: writing the comment mail fails → the comment is still saved, only the mail is dropped.
Postconditions	Mail in the recipient's Inbox; a system mail can only be forwarded (FR-MAIL-21)

Field validations

Field	Rule	Error message
Delete forever	Mail must be in Trash or be a draft	"Move the mail to Trash before deleting it forever."
Mail id (open / change)	Must be in the caller's mailbox	"Could not open this mail." / "Could not update the mail."
Attachment (download)	Caller uploaded it or holds the mail	"Could not download the file."

Business rules applied: BR-MAIL-02, BR-MAIL-03, BR-MAIL-04, BR-MAIL-05, BR-MAIL-11, BR-MAIL-12, BR-MAIL-13; global rules on authorised access and private messages (overall BRD §6).

Notifications / side effects

Event	Who is notified	Channel
Mail opened	– (counts and menu badge decrease)	In-app (Mail)
Comment added / edited / deleted	Task assignee, recipients, acting person	Mail from "TaskFlow" (if toggle on); in-app per Notification FRD
Task reassigned	Task's active participants	Mail from the acting person; in-app per Notification FRD
Worked hours changed	Task's active assignee and recipients	Mail from the acting person; in-app per Notification FRD

Messages

Situation	Message
Trash actions	"Moved to Trash" · "Restored" · "Deleted forever"
Delete forever outside Trash	"Move the mail to Trash before deleting it forever."
Reply to a system mail	"Mails from TaskFlow cannot be replied to. Forward it instead."
Empty folder / no search hits / nothing selected	"Nothing in *folder*." · "No mail matches your search." · "Select a mail to read it."
Failures	"Could not load your mail." · "Could not open this mail." · "Could not update the mail." · "Could not mark as unread." · "Could not download the file."
No permission to open the screen	Returned to the dashboard; no message
No permission for an action	The related "Could not …" message

4.1.2 Screen: Compose Panel (inside Mail)

Field	Value
Route / location	Side drawer on `/mail`, titled "New mail", "Reply", "Reply all", "Forward" or "Draft"
Access	`mail.compose`; Attach, chip removal and uploads need `mail.attachments`; Discard of a saved draft needs `mail.organize` (open issue §10)
Entry points	Compose button; Reply / Reply all / Forward in the reading pane; clicking a mail in Drafts; Assistant mail draft confirmed by the user (Assistant FRD)

Screen elements

Element	Type	Description	Mandatory	Visible to
To	Recipient picker (chips)	Names or emails; suggestions while typing; invalid chips marked "Not a TaskFlow user"	Yes (on send)	`mail.compose`
Cc	Recipient picker (chips)	Shown after clicking Cc	No	`mail.compose`
Subject	Text input	At most 200 characters	Subject or Message	`mail.compose`
Message	Multi-line text	Plain text, at most 20,000 characters	Subject or Message	`mail.compose`
Attach	Button (file picker)	Adds files, "Uploading…" while in progress	No	Shown to every composer; use needs `mail.attachments`
File chips	Chips with remove (x)	Attached and forwarded files	–	Same as Attach
Discard	Button	Closes the panel, removes uploads; deletes a saved draft	–	`mail.compose`
Save draft	Button	Saves to Drafts	–	`mail.compose`
Send	Button	Validates and sends	–	`mail.compose`

Functional requirements

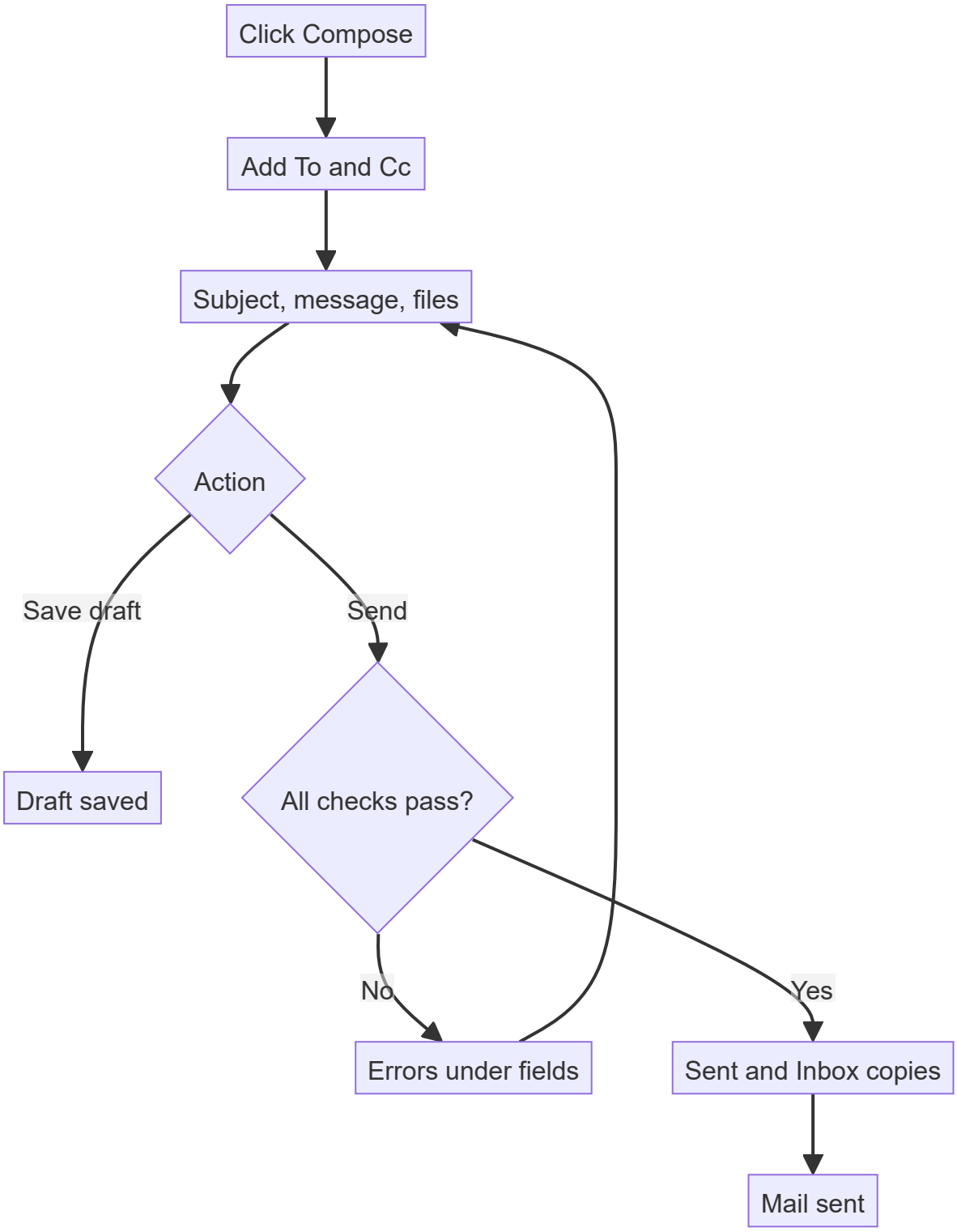
ID	Requirement	Priority	Permission	BRD trace
FR-MAIL-31	The system shall show Compose, Reply, Reply all and Forward only to users holding `mail.compose`.	Must	`mail.compose`	REQ-MAIL-28
FR-MAIL-32	The system shall title the panel "New mail", "Reply", "Reply all", "Forward" or "Draft" according to how it was opened.	Should	`mail.compose`	REQ-MAIL-28
FR-MAIL-33	The system shall suggest, while the user types in To or Cc, at most 10 active users whose name or email contains the typed text.	Should	`mail.compose`	REQ-MAIL-29
FR-MAIL-34	The system shall check each entered recipient, show "Checking recipients…" while checking, and mark any entry that is not an active TaskFlow user as "Not a TaskFlow user" with "Not a TaskFlow user: *addresses*. Mail can only go to people in TaskFlow."	Must	`mail.compose`	REQ-MAIL-30, BR-MAIL-01
FR-MAIL-35	The system shall refuse the whole send, delivering to nobody, when any recipient is not an active TaskFlow user, with "Remove or fix recipients that are not TaskFlow users." in the panel or "Not an active TaskFlow user: *addresses*." under To or Cc.	Must	`mail.compose`	REQ-MAIL-31, BR-MAIL-01, NFR-MAIL-06
FR-MAIL-36	The system shall require between 1 and 50 recipients across To and Cc, with "Add at least one recipient." for none and "A mail can have at most 50 recipients." / "Add between 1 and 50 recipients." for more than 50.	Must	`mail.compose`	REQ-MAIL-32, BR-MAIL-06
FR-MAIL-37	The system shall refuse a recipient address that is not in email form (needs "@", no spaces, at most 320 characters) with "'*address*' is not an email address."	Must	`mail.compose`	REQ-MAIL-30, REQ-MAIL-32
FR-MAIL-38	The system shall keep a person who is listed in both To and Cc in To only and count that person once.	Should	`mail.compose`	REQ-MAIL-33, BR-MAIL-06
FR-MAIL-39	The system shall refuse to send a mail that has neither subject nor message, with "Write a subject or a message before sending."	Must	`mail.compose`	REQ-MAIL-34, BR-MAIL-07
FR-MAIL-40	The system shall refuse a subject longer than 200 characters and a message longer than 20,000 characters.	Must	`mail.compose`	REQ-MAIL-35, BR-MAIL-07
FR-MAIL-41	The system shall, on a valid send, place the mail in the sender's Sent folder and in the Inbox of every To and Cc recipient, close the panel and show "Mail sent".	Must	`mail.compose`	REQ-MAIL-36
FR-MAIL-42	The system shall pre-fill Reply with the original sender as To (or the original recipients when the user sent the original), the subject prefixed "Re: " and the original text quoted under "On *dd MMM yyyy, HH:mm*, *name* wrote:".	Must	`mail.compose`	REQ-MAIL-37
FR-MAIL-43	The system shall pre-fill Reply all with the sender and everyone on the original To and Cc, leaving out the user.	Must	`mail.compose`	REQ-MAIL-38
FR-MAIL-44	The system shall pre-fill Forward with no recipients, the subject prefixed "Fwd: ", a "---------- Forwarded message ---------" header with the original text, and the original attachments as chips.	Must	`mail.compose`, `mail.attachments`	REQ-MAIL-39

ID	Requirement	Priority	Permission	BRD trace
FR-MAIL-45	The system shall add a sent reply or forward to the conversation of the original mail.	Should	`mail.compose`	REQ-MAIL-40
FR-MAIL-46	The system shall write dates in quoted and forwarded text in the application timezone and cut the pre-filled message at 20,000 characters.	Should	`mail.compose`	REQ-MAIL-37, REQ-MAIL-39
FR-MAIL-47	The system shall save a draft on Save draft, even with missing or invalid recipients, show "Draft saved", and check recipients only when the draft is sent.	Must	`mail.compose`	REQ-MAIL-41, BR-MAIL-10
FR-MAIL-48	The system shall reopen a draft from Drafts in the panel titled "Draft" with its recipients, subject, message and attachments, and on Send apply FR-MAIL-34 – FR-MAIL-41 and remove it from Drafts.	Must	`mail.compose`	REQ-MAIL-42
FR-MAIL-49	The system shall, on Discard, close the panel, remove the files uploaded in this panel and, for a saved draft, delete the draft, showing "Draft discarded".	Should	`mail.compose`, `mail.organize`	REQ-MAIL-43
FR-MAIL-50	The system shall allow at most 5 attachments on one mail and refuse more with "A mail can have at most 5 attachments."	Must	`mail.attachments`	REQ-MAIL-44, BR-MAIL-08
FR-MAIL-51	The system shall refuse a file larger than 10 MB ("*file* is larger than 10 MB." / "A file can be at most 10 MB.") and attachments totalling more than 25 MB on one mail, counting forwarded files ("Attachments on one mail can total at most 25 MB.").	Must	`mail.attachments`	REQ-MAIL-45, BR-MAIL-08
FR-MAIL-52	The system shall refuse empty files ("*file* is empty." / "The file is empty.") and files of type `.exe`, `.dll`, `.bat`, `.cmd`, `.com`, `.msi`, `.ps1`, `.sh`, `.js`, `.vbs` and `.scr` ("Files of type *.ext* are not allowed."), checking both in the browser and on the server.	Must	`mail.attachments`	REQ-MAIL-46, BR-MAIL-08, NFR-MAIL-03
FR-MAIL-53	The system shall refuse a new upload when the user already has 20 pending uploads ("You have *N* files waiting to be attached. Send or discard a mail before uploading more.") or when pending uploads reach 100 MB ("Your files waiting to be attached already total 100 MB. Send or discard a mail before uploading more.").	Should	`mail.attachments`	REQ-MAIL-47, BR-MAIL-09
FR-MAIL-54	The system shall let the user remove an attachment chip; a removed own upload that is on no mail is deleted.	Should	`mail.attachments`	REQ-MAIL-43, REQ-MAIL-44
FR-MAIL-55	The system shall accept on a mail only attachments uploaded by the sender that are not already on another mail, and otherwise refuse the send with "One or more attachments are not available."	Must	`mail.attachments`	REQ-MAIL-44, BR-MAIL-11
FR-MAIL-56	The system shall show each validation error under the field that caused it (To, Cc, Message or the files) and keep the panel open with the entered content.	Must	`mail.compose`	REQ-MAIL-31 – REQ-MAIL-35, NFR-MAIL-04

Use case: UC-MAIL-06 – Compose and send a mail

Item	Detail
Actor	User with `mail.compose`

Item	Detail
Preconditions	Mail screen open
Trigger	Clicks Compose
Main flow	1. Panel "New mail" opens (FR-MAIL-32). 2. User types in To (and Cc); suggestions appear and entries are checked (FR-MAIL-33, 34). 3. User enters subject and/or message, may attach files (UC-MAIL-09). 4. User clicks Send. 5. System validates recipients, content and attachments (FR-MAIL-35 – 40, 50 – 55). 6. System stores Sent and Inbox copies and shows "Mail sent" (FR-MAIL-41).
Alternate flows	A1: Save draft → UC-MAIL-08. A2: mail started by the Assistant → the user reviews and confirms (Assistant FRD).
Exceptions	E1: non-user recipient → whole send refused (FR-MAIL-35). E2: 0 or more than 50 recipients (FR-MAIL-36). E3: no subject and no message (FR-MAIL-39). E4: send fails → "Could not send the mail."
Postconditions	Sender's Sent and each recipient's Inbox hold the mail; recipients' badges rise within 15 seconds



Use case: UC-MAIL-07 – Reply, reply all or forward

Item	Detail
Actor	User with `mail.compose`
Preconditions	A mail open in the reading pane
Trigger	Clicks Reply, Reply all or Forward
Main flow	1. System pre-fills the panel (FR-MAIL-42 – 44, 46). 2. User edits and clicks Send. 3. Validation and delivery as UC-MAIL-06. 4. The new mail joins the conversation (FR-MAIL-45).

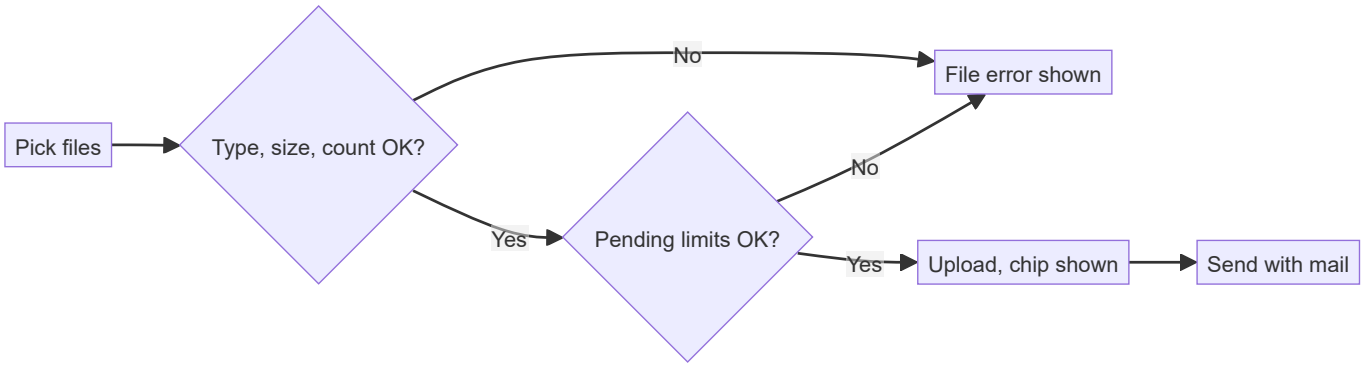
Item	Detail
Alternate flows	A1: system mail → only Forward offered (FR-MAIL-21). A2: mail in Trash → only Forward offered (FR-MAIL-22).
Exceptions	E1: reply requested for a system mail → "Mails from TaskFlow cannot be replied to. Forward it instead." E2: pre-fill fails → "Could not prepare the reply."
Postconditions	Reply or forward delivered; conversation extended

Use case: UC-MAIL-08 – Save, continue or discard a draft

Item	Detail
Actor	User with `mail.compose` (`mail.organize` to discard a saved draft)
Preconditions	Compose panel open, or a draft in Drafts
Trigger	Clicks Save draft, opens a draft from Drafts, or clicks Discard
Main flow	1. Save draft → stored with whatever recipients are present; "Draft saved" (FR-MAIL-47). 2. Later the user opens it from Drafts (FR-MAIL-48). 3. User clicks Send → full checks; draft leaves Drafts.
Alternate flows	A1: Discard → panel closes, panel uploads removed, saved draft deleted; "Draft discarded" (FR-MAIL-49).
Exceptions	E1: "Could not save the draft." · "Could not open the draft." · "Could not discard the draft."
Postconditions	Draft saved, sent or removed

Use case: UC-MAIL-09 – Attach files

Item	Detail
Actor	User with `mail.attachments`
Preconditions	Compose panel open
Trigger	Clicks Attach and picks files
Main flow	1. System checks type, emptiness, 10 MB per file, 5 files and 25 MB total (FR-MAIL-50 – 52). 2. System checks the pending-upload limits (FR-MAIL-53). 3. Accepted files upload ("Uploading…") and appear as chips. 4. On send the files go with the mail (FR-MAIL-55).
Alternate flows	A1: user removes a chip (FR-MAIL-54). A2: Forward carries the original files (FR-MAIL-44).
Exceptions	E1: a limit is broken → the file's message under the files. E2: upload fails → "Could not upload *file*." E3: no file chosen → "Choose a file to upload."
Postconditions	Files attached to the mail or draft



Field validations

Field	Rule	Error message
To / Cc	At least one recipient after removing duplicates	"Add at least one recipient."
To / Cc	At most 50 recipients	"A mail can have at most 50 recipients." / "Add between 1 and 50 recipients."
To / Cc	Email form, at most 320 characters	"'*address*' is not an email address."
To / Cc	Every recipient is an active TaskFlow user (on send)	"Not an active TaskFlow user: *addresses*." / "Remove or fix recipients that are not TaskFlow users."
To / Cc	Checking in progress	"Checking recipients…"
Subject / Message	At least one present (on send)	"Write a subject or a message before sending."
Subject	At most 200 characters	Maximum-length message under the field
Message	At most 20,000 characters	Maximum-length message under the field
Attachments	At most 5	"A mail can have at most 5 attachments."
Attachments	Total at most 25 MB	"Attachments on one mail can total at most 25 MB."
File	At most 10 MB	"*file* is larger than 10 MB." / "A file can be at most 10 MB."
File	Not empty	"*file* is empty." / "The file is empty."
File	Type not blocked	"Files of type *.ext* are not allowed."
File	A file chosen	"Choose a file to upload."
Pending uploads	At most 20 files and under 100 MB	"You have *N* files waiting to be attached. Send or discard a mail before uploading more." / "Your files waiting to be attached already total 100 MB. Send or discard a mail before uploading more."
Attachments	Own and unused	"One or more attachments are not available."

Business rules applied: BR-MAIL-01, BR-MAIL-05, BR-MAIL-06, BR-MAIL-07, BR-MAIL-08, BR-MAIL-09, BR-MAIL-10, BR-MAIL-11.

Notifications / side effects

Event	Who is notified	Channel
Mail sent	Every To and Cc recipient	Mail (Inbox copy; menu badge)

Messages

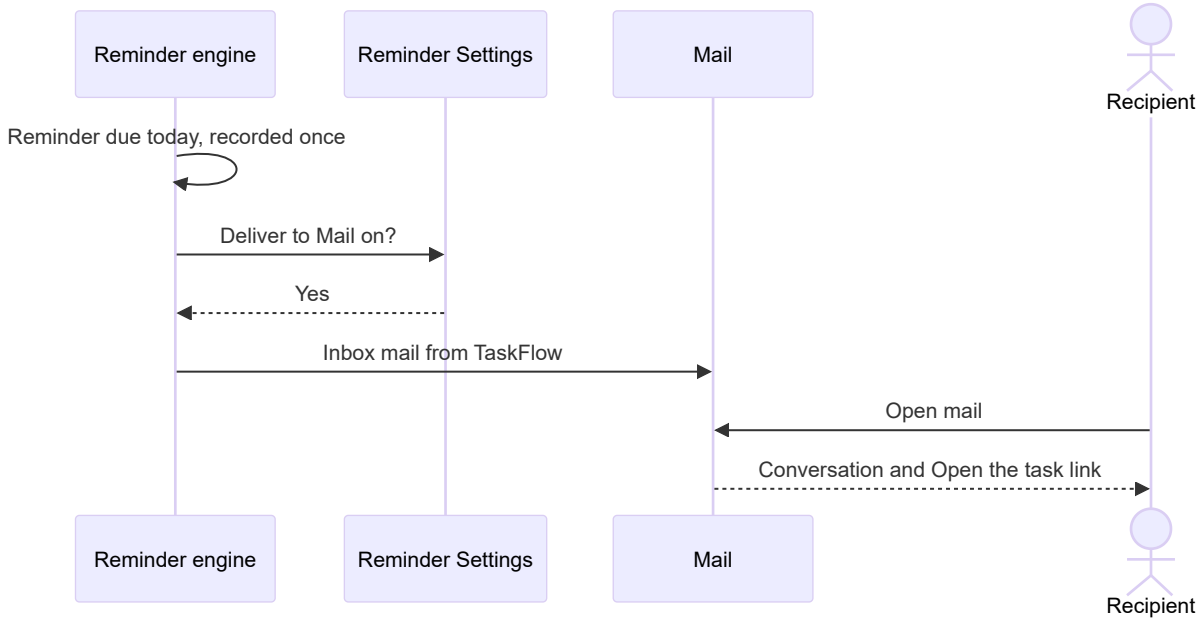
Situation	Message
Success	"Mail sent" · "Draft saved" · "Draft discarded"
Failures	"Could not send the mail." · "Could not save the draft." · "Could not prepare the reply." · "Could not open the draft." · "Could not discard the draft." · "Could not upload *file*."
No permission	Compose actions are not shown; a refused action shows the related "Could not …" message

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5. Cross-Module Workflows

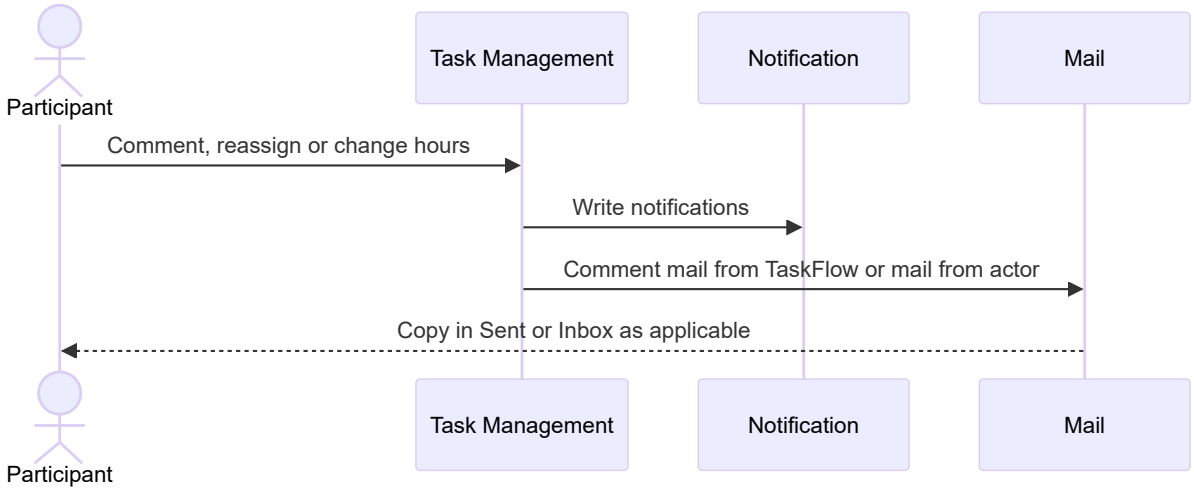
5.1 Deadline reminder delivered to the Inbox

Step	Module	Actor / System	Action	FR reference
1	NOTIF	System (reminder engine)	Finds a reminder due today and records it	FR-NOTIF-32 – FR-NOTIF-44, FR-NOTIF-48 (Notification FRD)
2	NOTIF	System	Writes the in-app notification	FR-NOTIF-45, FR-NOTIF-46
3	REM	System	Reads "Also deliver reminders to Mail"	Reminder Settings FRD
4	MAIL	System	If on, places an Inbox mail from "TaskFlow" ("Task reminder:", "Task due today:" or "Overdue task:" + title) with an "Open the task" link	FR-NOTIF-47, FR-MAIL-12, FR-MAIL-21
5	MAIL	Recipient	Opens the mail, opens the task or forwards it	FR-MAIL-09, FR-MAIL-12, FR-MAIL-44



5.2 Task comment, reassignment or worked hours

Step	Module	Actor / System	Action	FR reference
1	TASK	Participant	Adds, edits or deletes a comment; reassigns a task; changes worked hours	Task Management FRD
2	NOTIF	System	Writes in-app notifications	FR-NOTIF-23 – FR-NOTIF-26
3	MAIL	System	Comment → mail from "TaskFlow" (if toggle on); reassignment / worked hours → mail from the acting person	FR-MAIL-25, FR-MAIL-26, FR-MAIL-27
4	MAIL	Recipient	Reads the mail; replies only to mails from a person	FR-MAIL-21



> If writing the comment mail fails, the comment is still saved and only the mail is dropped (BR-NOTIF-13; Mail BRD §4.5).

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6. Reports & Exports

Not applicable for this sub-module.

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7. Notifications & Messages Catalogue

Code	Trigger	Recipients	Channel	Text
M-MAIL-01	User sends a mail	To and Cc recipients	Mail (Inbox)	Subject and message as written by the sender
M-MAIL-02	First or second reminder delivered (toggle on)	Reminder recipients	Mail from "TaskFlow"	Subject "Task reminder: *title*"; body = reminder text, "Open the task" link, "This is an automatic reminder from TaskFlow."
M-MAIL-03	Due-today reminder delivered (toggle on)	Reminder recipients	Mail from "TaskFlow"	Subject "Task due today: *title*"; body as M-MAIL-02
M-MAIL-04	Overdue reminder delivered (toggle on)	Reminder recipients	Mail from "TaskFlow"	Subject "Overdue task: *title*"; body as M-MAIL-02

Code	Trigger	Recipients	Channel	Text
M-MAIL-05	Comment added / edited / deleted (toggle on)	Assignee, recipients, acting person	Mail from "TaskFlow"	Subject "Comment *verb*: *title*"
M-MAIL-06	Task reassigned	Task's active participants	Mail from the acting person	Subject "Task reassigned: *title*"
M-MAIL-07	Worked hours changed	Task's active assignee and recipients	Mail from the acting person	Subject "Worked hours changed: *title*"

User messages are listed per screen in §4.1.1 and §4.1.2.

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8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Mail message	One written mail or draft: subject, message, sender (none for TaskFlow), system flag, related task, related reminder, conversation, previous mail	User (send, draft); System (system mails)	Subject ≤ 200, message ≤ 20,000 characters; a draft has no sent time
Mail recipient	A user a sent mail is addressed to, as To or Cc	System on send	One entry per user per mail; To wins over Cc
Mailbox copy	One user's entry for a mail in Inbox, Sent or Drafts with read, starred, important, trashed and deleted-forever state	System on send / draft	Personal to the owner; delete forever hides the copy, it is not physically removed
Mail attachment	A file with name, type, size and uploader; on no mail while pending	User (upload)	Limits per BR-MAIL-08 and BR-MAIL-09; files kept even after delete forever; stored outside the database (NFR-MAIL-08)

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9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-MAIL-01	25 mails per page	Message list (FR-MAIL-05)
NFR-MAIL-02	Every action works only on the caller's mailbox; foreign mail reported as not found	All actions (FR-MAIL-20, FR-MAIL-28)
NFR-MAIL-03	Files always downloaded; blocked types refused; plain text with only web links clickable	Reading pane, attachments (FR-MAIL-11, FR-MAIL-29, FR-MAIL-52)
NFR-MAIL-04	Errors shown under the field that caused them	Compose panel (FR-MAIL-56)
NFR-MAIL-05	One pane at a time on narrow screens	Mail screen (FR-MAIL-24)
NFR-MAIL-06	A send is all or nothing	Send (FR-MAIL-35, FR-MAIL-55)
NFR-MAIL-07	Menu badge refreshed on page change and every 15 seconds	Menu badge (FR-MAIL-04)

ID	Requirement	Affects
NFR-MAIL-08	Attachment files stored outside the database and included in backups	Attachments (§8)
NFR-MAIL-09	Access controlled by the four `mail.*` permissions, editable in Role Configuration	All screens (§3.2)

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	All correspondents are TaskFlow users of one organisation; plain text is sufficient.	Jojo George
Assumption	Recipient suggestions over all active users are acceptable for a single organisation (BRD R-MAIL-02).	Jojo George
Open issue	⚠️ To be confirmed: Attach, chip removal and download buttons are shown to every composer / reader even without `mail.attachments`; such users get the "Could not …" error. Confirm whether they should be hidden.	Jojo George
Open issue	⚠️ To be confirmed: Discard of a saved draft needs `mail.organize`; a role with `mail.compose` only cannot discard a saved draft. Confirm whether intended.	Jojo George
Open issue	⚠️ To be confirmed: reassignment and worked-hours mails are sent even when "Also deliver reminders to Mail" is off. Confirm whether they should follow the toggle.	Jojo George
Open issue	⚠️ To be confirmed: original Screen 14 FR 6 / FR 7 (record delivery failures) – with internal Mail the notification and the Inbox mail are written together; confirm whether a separate failure record is needed.	Jojo George
Open issue	Deleted mail and attachment files are never physically removed and abandoned uploads are not cleaned up (future phase, BRD R-MAIL-01).	Jojo George

11. Requirements Traceability Matrix

Test case column: automated test suites from the TaskFlow test projects (backend test classes and frontend `*.spec.ts` files). The mapping was checked against the test sources; per-test mapping belongs in the LLD. "Manual UI check" = no automated test covers the behaviour.

BRD requirements → FRD

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-MAIL-01	FR-MAIL-01	Mail	UC-MAIL-01	`MailEndpointsTests`
REQ-MAIL-02	FR-MAIL-02	Mail	UC-MAIL-01	`MailboxTests`
REQ-MAIL-03	FR-MAIL-03	Mail	UC-MAIL-01	`MailboxTests`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-MAIL-04	FR-MAIL-04	Mail (menu)	UC-MAIL-01	`MailEndpointsTests`, `shell.component.spec.ts`
REQ-MAIL-05	FR-MAIL-05, FR-MAIL-08	Mail	UC-MAIL-01	`MailboxTests`, `mail-list.component.spec.ts`
REQ-MAIL-06	FR-MAIL-06	Mail	UC-MAIL-01	`MailboxTests`, `mail-list.component.spec.ts`
REQ-MAIL-07	FR-MAIL-07, FR-MAIL-08	Mail	UC-MAIL-01	`MailboxTests`
REQ-MAIL-08	FR-MAIL-09, FR-MAIL-30	Mail	UC-MAIL-01	`MailDraftAndThreadTests`
REQ-MAIL-09	FR-MAIL-10	Mail	UC-MAIL-01	`MailboxTests`
REQ-MAIL-10	FR-MAIL-11	Mail	UC-MAIL-01	`linkify.spec.ts`, `mail-reader.component.spec.ts`
REQ-MAIL-11	FR-MAIL-12	Mail	UC-MAIL-01, UC-MAIL-05	`mail-reader.component.spec.ts`
REQ-MAIL-12	FR-MAIL-13, FR-MAIL-30	Mail	UC-MAIL-02	`MailboxTests`
REQ-MAIL-13	FR-MAIL-14, FR-MAIL-30	Mail	UC-MAIL-02	`MailboxTests`
REQ-MAIL-14	FR-MAIL-15, FR-MAIL-30	Mail	UC-MAIL-02	`MailboxTests`
REQ-MAIL-15	FR-MAIL-16, FR-MAIL-17, FR-MAIL-30	Mail	UC-MAIL-03	`MailboxTests`
REQ-MAIL-16	FR-MAIL-18	Mail	UC-MAIL-03	`MailboxTests`
REQ-MAIL-17	FR-MAIL-19	Mail	UC-MAIL-02, UC-MAIL-03	`MailboxTests`
REQ-MAIL-18	FR-MAIL-20	Mail	UC-MAIL-01	`MailboxTests`, `MailEndpointsTests`
REQ-MAIL-19	FR-MAIL-21	Mail	UC-MAIL-05, UC-MAIL-07	`MailDraftAndThreadTests`
REQ-MAIL-20	FR-MAIL-22	Mail	UC-MAIL-07	`mail-reader.component.spec.ts`
REQ-MAIL-21	FR-MAIL-23	Mail	UC-MAIL-01	Manual UI check
REQ-MAIL-22	FR-MAIL-24	Mail	UC-MAIL-01	`mail.component.spec.ts`
REQ-MAIL-23	FR-MAIL-25	Mail	UC-MAIL-05	`TaskCommentActivityNotificationTests`
REQ-MAIL-24	FR-MAIL-26	Mail	UC-MAIL-05	`TaskServiceTests`
REQ-MAIL-25	FR-MAIL-27	Mail	UC-MAIL-05	`TaskWorkedHoursNotificationTests`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-MAIL-26	FR-MAIL-28	Mail	UC-MAIL-04	`MailAttachmentTests`
REQ-MAIL-27	FR-MAIL-29	Mail	UC-MAIL-04	`MailEndpointsTests`
REQ-MAIL-28	FR-MAIL-31, FR-MAIL-32	Compose panel	UC-MAIL-06	`MailEndpointsTests`, `compose-dialog.component.spec.ts`
REQ-MAIL-29	FR-MAIL-33	Compose panel	UC-MAIL-06	`MailboxTests`, `MailEndpointsTests`, `recipient-picker.component.spec.ts`
REQ-MAIL-30	FR-MAIL-34, FR-MAIL-37	Compose panel	UC-MAIL-06	`MailSendTests`, `recipient-picker.component.spec.ts`
REQ-MAIL-31	FR-MAIL-35, FR-MAIL-56	Compose panel	UC-MAIL-06	`MailSendTests`
REQ-MAIL-32	FR-MAIL-36, FR-MAIL-37, FR-MAIL-56	Compose panel	UC-MAIL-06	`MailSendTests`
REQ-MAIL-33	FR-MAIL-38	Compose panel	UC-MAIL-06	`MailSendTests`
REQ-MAIL-34	FR-MAIL-39, FR-MAIL-56	Compose panel	UC-MAIL-06	`MailSendTests`
REQ-MAIL-35	FR-MAIL-40, FR-MAIL-56	Compose panel	UC-MAIL-06	`MailSendTests`
REQ-MAIL-36	FR-MAIL-41	Compose panel	UC-MAIL-06	`MailSendTests`
REQ-MAIL-37	FR-MAIL-42, FR-MAIL-46	Compose panel	UC-MAIL-07	`MailDraftAndThreadTests`
REQ-MAIL-38	FR-MAIL-43	Compose panel	UC-MAIL-07	`MailDraftAndThreadTests`
REQ-MAIL-39	FR-MAIL-44, FR-MAIL-46	Compose panel	UC-MAIL-07	`MailDraftAndThreadTests`
REQ-MAIL-40	FR-MAIL-45	Compose panel	UC-MAIL-07	`MailDraftAndThreadTests`
REQ-MAIL-41	FR-MAIL-47	Compose panel	UC-MAIL-08	`MailDraftAndThreadTests`
REQ-MAIL-42	FR-MAIL-48	Compose panel	UC-MAIL-08	`MailDraftAndThreadTests`
REQ-MAIL-43	FR-MAIL-49, FR-MAIL-54	Compose panel	UC-MAIL-08, UC-MAIL-09	`MailAttachmentTests`, `compose-dialog.component.spec.ts`
REQ-MAIL-44	FR-MAIL-50, FR-MAIL-54, FR-MAIL-55	Compose panel	UC-MAIL-09	`MailAttachmentTests`
REQ-MAIL-45	FR-MAIL-51	Compose panel	UC-MAIL-09	`MailAttachmentTests`
REQ-MAIL-46	FR-MAIL-52	Compose panel	UC-MAIL-09	`MailAttachmentTests`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-MAIL-47	FR-MAIL-53	Compose panel	UC-MAIL-09	`MailAttachmentTests`

BRD business rules and NFRs → FRD

BRD item	FRD requirement(s)	Screen	Use case	Test case
BR-MAIL-01	FR-MAIL-34, FR-MAIL-35	Compose panel	UC-MAIL-06	`MailSendTests`
BR-MAIL-02	FR-MAIL-20	Mail	UC-MAIL-01	`MailboxTests`
BR-MAIL-03	FR-MAIL-19	Mail	UC-MAIL-02	`MailboxTests`
BR-MAIL-04	FR-MAIL-18	Mail	UC-MAIL-03	`MailboxTests`
BR-MAIL-05	FR-MAIL-21	Mail, Compose panel	UC-MAIL-07	`MailDraftAndThreadTests`
BR-MAIL-06	FR-MAIL-36, FR-MAIL-38	Compose panel	UC-MAIL-06	`MailSendTests`
BR-MAIL-07	FR-MAIL-39, FR-MAIL-40	Compose panel	UC-MAIL-06	`MailSendTests`
BR-MAIL-08	FR-MAIL-50, FR-MAIL-51, FR-MAIL-52	Compose panel	UC-MAIL-09	`MailAttachmentTests`
BR-MAIL-09	FR-MAIL-53	Compose panel	UC-MAIL-09	`MailAttachmentTests`
BR-MAIL-10	FR-MAIL-47	Compose panel	UC-MAIL-08	`MailDraftAndThreadTests`
BR-MAIL-11	FR-MAIL-28, FR-MAIL-55	Mail, Compose panel	UC-MAIL-04	`MailAttachmentTests`
BR-MAIL-12	FR-MAIL-23	Mail	UC-MAIL-01	Manual UI check
BR-MAIL-13	FR-MAIL-25, FR-MAIL-26, FR-MAIL-27	Mail	UC-MAIL-05	`TaskCommentActivityNotificationTests`, `TaskServiceTests`, `TaskWorkedHoursNotificationTests`

BRD item	FRD requirement(s)	Screen	Use case	Test case
NFR-MAIL-01	FR-MAIL-05	Mail	UC-MAIL-01	`MailboxTests`
NFR-MAIL-02	FR-MAIL-20, FR-MAIL-28	Mail	UC-MAIL-01, UC-MAIL-04	`MailEndpointsTests`
NFR-MAIL-03	FR-MAIL-11, FR-MAIL-29, FR-MAIL-52	Mail, Compose panel	UC-MAIL-04, UC-MAIL-09	`MailAttachmentTests`, `MailEndpointsTests`, `linkify.spec.ts`, `mail-reader.component.spec.ts`
NFR-MAIL-04	FR-MAIL-56	Compose panel	UC-MAIL-06	`compose-dialog.component.spec.ts`
NFR-MAIL-05	FR-MAIL-24	Mail	UC-MAIL-01	`mail.component.spec.ts`
NFR-MAIL-06	FR-MAIL-35, FR-MAIL-55	Compose panel	UC-MAIL-06	`MailSendTests`
NFR-MAIL-07	FR-MAIL-04	Mail (menu)	UC-MAIL-01	`shell.component.spec.ts`, `mail.service.spec.ts`
NFR-MAIL-08	§8 Mail attachment	Mail, Compose panel	UC-MAIL-09	– (deployment / backup check)
NFR-MAIL-09	FR-MAIL-01, FR-MAIL-13 – FR-MAIL-18, FR-MAIL-28, FR-MAIL-31 (§3.2)	Mail, Compose panel	All	`MailEndpointsTests`

BRD acceptance criteria → FRD

Acceptance criterion	FRD requirement(s)
AC-MAIL-01	FR-MAIL-01, FR-MAIL-02, FR-MAIL-03
AC-MAIL-02	FR-MAIL-04
AC-MAIL-03	FR-MAIL-05, FR-MAIL-06, FR-MAIL-07
AC-MAIL-04	FR-MAIL-09, FR-MAIL-10, FR-MAIL-11, FR-MAIL-12
AC-MAIL-05	FR-MAIL-13 – FR-MAIL-19
AC-MAIL-06	FR-MAIL-20
AC-MAIL-07	FR-MAIL-21, FR-MAIL-22
AC-MAIL-08	FR-MAIL-23, FR-MAIL-24
AC-MAIL-09	FR-MAIL-25, FR-MAIL-26, FR-MAIL-27
AC-MAIL-10	FR-MAIL-28, FR-MAIL-29
AC-MAIL-11	FR-MAIL-31 – FR-MAIL-35
AC-MAIL-12	FR-MAIL-36 – FR-MAIL-40
AC-MAIL-13	FR-MAIL-41

Acceptance criterion	FRD requirement(s)
AC-MAIL-14	FR-MAIL-42 – FR-MAIL-46
AC-MAIL-15	FR-MAIL-47, FR-MAIL-48, FR-MAIL-49
AC-MAIL-16	FR-MAIL-50 – FR-MAIL-53

Untraced BRD items: 0.

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	To be assigned	Pending	–

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